Concept 1 | Fundamentals of Calculus (Limits and Derivatives)

English Student Edition • Focused formulas and guided practice

Formula opener

Derivative

$$\frac{d}{dx}x^n = nx^{n-1}$$

Product

$$(uv)' = u'v + uv'$$

Chain

$$\frac{dy}{dx} = \frac{dy}{du}\frac{du}{dx}$$

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Worked Solutions **Fundamentals of Calculus (Limits and Derivatives)**

Q001 Written

For $f(x) = x^2 + 2x$, find the average rate of change from -1 to 3, and from 3 to $3 + h$.

Solution

1. Use $[f(b) - f(a)]/(b - a)$. For the h-form, factor $f(3 + h) - f(3) = h(8 + h)$.

Final answer

$4; \quad 8 + h$

Q002 Written

Find the average rate of change for the three given Try functions on their stated intervals.

Solution

1. Evaluate both endpoint values, subtract, and divide by the change in x; cancel h in the variable-endpoint case.

Final answer

$2; \quad -5/3; \quad h - 6$

Worked Solutions **Fundamentals of Calculus (Limits and Derivatives)**

Q003 Written

Find the average rate of change for all nine given Exercise functions, including fixed and h-form intervals.

Solution

1. Keep the endpoint order consistent and simplify before substituting numerical values.

Final answer

Apply $[f(b) - f(a)]/(b - a)$ to each printed item exactly.

Q004 Written

Find the average rate of change of a circle's area as its diameter changes from 10 cm to 18 cm.

Solution

1. With $A = \pi d^2/4$, compute $[A(18) - A(10)]/(18 - 10) = 7\pi$.

Final answer

$7\pi\ \text{cm}^2/\text{cm}$

Worked Solutions **Fundamentals of Calculus (Limits and Derivatives)**

Q005 Written

Find the given limits by direct substitution and by simplifying the indeterminate fraction before substitution.

Solution

1. Direct substitution is valid when defined; otherwise factor/cancel first, then put the limiting value into the simplified expression.

Final answer

-2; 6

Q006 Written

Find the three given Try limits, including polynomial and h-quotient forms.

Solution

1. Test direct substitution, identify 0/0, simplify, and substitute again.

Final answer

Substitute after removing any common factor; polynomial limits are obtained directly.

Worked Solutions

Fundamentals of Calculus (Limits and Derivatives)

Q007 Written

Find all nine given Exercise limits, including x-limits and h-limits.

Solution

1. For a polynomial use continuity; for a quotient, factor before taking the limit.

Final answer

Each limit is obtained by direct substitution or cancellation of the factor causing 0/0.

Q008 Written

If $\lim_{x\to 3}(x^2 - 5x + k)/(x - 3) = L$ is finite, find Lk.

Solution

1. Set the numerator to zero at 3: $k = 6$. Then cancel $x - 3$, giving $L = 1$, so $Lk = 6$.

Final answer

6

Worked Solutions **Fundamentals of Calculus (Limits and Derivatives)**

Q009 Written

Find the differential coefficient of $f(x) = 3x^2$ at $x = -2$ using the definition.

Solution

1. Use $\lim_{h\to0}[f(-2+h) - f(-2)]/h = \lim(-12+3h) = -12$.

Final answer

$f'(-2) = -12$

Q010 Written

Find the differential coefficients at the given values for the two given Try functions using the definition.

Solution

1. Expand the shifted polynomial fully before cancelling the common h.

Final answer

Substitute a into $[f(a+h) - f(a)]/h$, cancel h, and let h tend to zero.

Worked Solutions **Fundamentals of Calculus (Limits and Derivatives)**

Q011 Written

Find the differential coefficients for all six given Exercise functions at their stated values.

Solution

1. The resulting constant is the instantaneous slope at the given input.

Final answer

Use the definition at each specified point and simplify exactly.

Q012 Written

A dropped stone has displacement $s(t) = 4.9t^2$ metres. Find its instantaneous velocity at $t = 10$ seconds using the definition.

Solution

1. The differential coefficient is $s'(t) = 9.8t$; hence $s'(10) = 98$.

Final answer

$98\ \mathrm{m\,s^{-1}}$

Worked Solutions **Fundamentals of Calculus (Limits and Derivatives)**

Q013 Written

Differentiate $f(x) = x^2 + 3x$ using the definition of the derivative.

Solution

1. Expand $f(x+h) - f(x) = 2xh + h^2 + 3h$, divide by h, and let h tend to zero.

Final answer

$f'(x) = 2x + 3$

Q014 Written

Differentiate the three given Try functions using the definition.

Solution

1. Do not use a memorised rule; retain the h-limit structure until the cancellation is visible.

Final answer

Expand each $f(x+h) - f(x)$, cancel h, and take the limit.

Worked Solutions **Fundamentals of Calculus (Limits and Derivatives)**

Q015 Written

Differentiate all nine given Exercise functions using the definition.

Solution

1. Expand, collect the coefficient of h, divide by h, then set $h = 0$.

Final answer

Apply $f'(x) = \lim_{h\to 0}[f(x+h) - f(x)]/h$ to every printed polynomial.

Q016 Written

A regular hexagon has side length x. Write its area and differentiate it using the definition.

Solution

1. Substitute the area formula into the difference quotient; the derivative of the quadratic factor is $3\sqrt{3}x$.

Final answer

$$A = \frac{3\sqrt{3}}{2}x^2, \quad A'(x) = 3\sqrt{3}\,x$$

Worked Solutions **Fundamentals of Calculus (Limits and Derivatives)**

Q017 Written

If $f(x) = 5$ for every x except that it is undefined at x=2, does $\lim_{x\to 2} f(x)$ exist?

Solution

1. A limit depends on nearby values, not on whether the function is defined at the point itself.

Final answer

Yes; the limit is 5.

Q018 MCQ

The average rate of change from a to b is:

Solution

1. Use the secant slope.

Correct option: B

Final answer

$[f(b) - f(a)]/(b - a)$

Worked Solutions **Fundamentals of Calculus (Limits and Derivatives)**

Q019 MCQ

For $f(x) = x^2 + 2x$ from -1 to 3, the average rate is:

Solution

1. Evaluate the two endpoints.

Correct option: C

Final answer

4

Q020 MCQ

The average area-rate for diameter 10 to 18 is:

Solution

1. Use $A = \pi d^2/4$.

Correct option: A

Final answer

7π

Worked Solutions **Fundamentals of Calculus (Limits and Derivatives)**

Q021 *MCQ*

The h-form average rate is found by:

Solution

1. Simplify the difference quotient.

Correct option: B

Final answer

cancelling the common h

Q022 *MCQ*

A direct-substitution limit is valid when:

Solution

1. Use continuity where defined.

Correct option: A

Final answer

the expression is defined

Worked Solutions **Fundamentals of Calculus (Limits and Derivatives)**

Q023 MCQ

The 0/0 form requires:

Solution

1. Remove the common vanishing factor.

Correct option: A

Final answer

factor/cancel first

Q024 MCQ

In the finite-limit challenge, k equals:

Solution

1. The numerator must vanish at $x = 3$.

Correct option: C

Final answer

6

Worked Solutions **Fundamentals of Calculus (Limits and Derivatives)**

Q025 MCQ

For a polynomial, the limit equals:

Solution

1. Polynomials are continuous.

Correct option: A

Final answer

its value at the point

Q026 MCQ

The definition of the differential coefficient at a is:

Solution

1. Use the difference quotient.

Correct option: A

Final answer

lim [f(a+h)-f(a)]/h

Worked Solutions **Fundamentals of Calculus (Limits and Derivatives)**

Q027 MCQ

For $f(x) = 3x^2$, $f'(-2)$ is:

Solution

1. Expand and cancel h.

Correct option: A

Final answer

-12

Q028 MCQ

The stone's velocity at 10 s is:

Solution

1. Use $s'(t) = 9.8t$.

Correct option: B

Final answer

98

Worked Solutions **Fundamentals of Calculus (Limits and Derivatives)**

Q029 MCQ

After cancelling h, take the limit as:

Solution

1. The definition uses h approaching zero.

Correct option: A

Final answer

$h \to 0$

Q030 MCQ

The derivative function is:

Solution

1. Associate the coefficient with each x.

Correct option: B

Final answer

the differential coefficient for every x

Worked Solutions **Fundamentals of Calculus (Limits and Derivatives)**

Q031 MCQ

The derivative of $x^2 + 3x$ from the definition is:

Solution

1. Expand the difference quotient.

Correct option: B

Final answer

$2x + 3$

Q032 MCQ

The area of a regular hexagon of side x is:

Solution

1. Use the standard hexagon area formula.

Correct option: B

Final answer

$(3\sqrt{3}/2)x^2$

Worked Solutions **Fundamentals of Calculus (Limits and Derivatives)**

Q033 MCQ

A function need not be defined at a for its limit to:

Solution

1. Nearby values determine a limit.

Correct option: A

Final answer

exist

Q034 MCQ

The derivative of a constant function is:

Solution

1. The difference quotient of a constant is zero.

Correct option: A

Final answer

0

Worked Solutions **Fundamentals of Calculus (Limits and Derivatives)**

Quiz MCQ 1 Concept Quiz · MCQ

Find the average rate of change of x^2 from 1 to 3.

Solution

1. $[9-1]/2 = 4$.

Correct option: A

Final answer

4

Quiz MCQ 2 Concept Quiz · MCQ

Find $\lim_{x\to 2}(x^2+1)$.

Solution

1. Direct substitution gives 5.

Correct option: B

Final answer

5

Worked Solutions **Fundamentals of Calculus (Limits and Derivatives)**

Quiz MCQ 3 **Concept Quiz · MCQ**

Find $f'(x)$ for $f(x) = x^2$ using the definition.

Solution

1. Expand $(x + h)^2 - x^2$, cancel h, and let h tend to zero.

Correct option: C

Final answer

$2x$

Quiz MCQ 4 **Concept Quiz · MCQ**

If $s(t) = 4.9t^2$, find the velocity at t=5.

Solution

1. $s'(t) = 9.8t$.

Correct option: D

Final answer

49

Worked Solutions **Fundamentals of Calculus (Limits and Derivatives)**

Quiz WRITTEN 5 **Concept Quiz · Written**

Find the average rate of change of $3x + 1$ from 2 to 7.

Solution

1. A linear function has constant slope 3.

Final answer

3

Quiz WRITTEN 6 **Concept Quiz · Written**

Evaluate $\lim_{h \to 0}(6 + 2h)$.

Solution

1. Substitute $h = 0$.

Final answer

6

Worked Solutions **Fundamentals of Calculus (Limits and Derivatives)**

Quiz WRITTEN 7 **Concept Quiz · Written**

Find the differential coefficient of $2x^2$ at x=3 from the definition.

Solution

1. The difference quotient simplifies to $4x + 2h$, giving 12.

Final answer

12

Quiz WRITTEN 8 **Concept Quiz · Written**

State $\lim_{x \to 2} f(x)$ when f(x)=5 for all x except x=2 where it is undefined.

Solution

1. Nearby values are constantly 5.

Final answer

5

Answer key

Use the question number shown on each page.

Q001 4; $8+h$

Q002 2; $-5/3$; $h-6$

Q003 Apply $[f(b)-f(a)]/(b-a)$ to each printed item exactly.

Q004 $7\pi\ \mathrm{cm}^2/\mathrm{cm}$

Q005 -2; 6

Q006 Substitute after removing any common factor; polynomial limits are obtained directly.

Q007 Each limit is obtained by direct substitution or cancellation of the factor causing 0/0.

Q008 6

Q009 $f'(-2)=-12$

Q010 Substitute a into $[f(a+h)-f(a)]/h$, cancel h, and let h tend to zero.

Q011 Use the definition at each specified point and simplify exactly.

Q012 $98\ \mathrm{m\,s^{-1}}$

Q013 $f'(x)=2x+3$

Q014 Expand each $f(x+h)-f(x)$, cancel h, and take the limit.

Q015 Apply $f'(x)=\lim_{h\to 0}[f(x+h)-f(x)]/h$ to every printed polynomial.

Q016 $A=\frac{3\sqrt{3}}{2}x^2$, $A'(x)=3\sqrt{3}\,x$

Q017 Yes; the limit is 5.

Q018 B

Q019 C

Q020 A

Q021 B

Q022 A

Q023 A

Q024 C

Q025 A

Q026 A

Q027 A

Q028 B

Q029 A

Q030 B

Q031 B

Q032 B

Q033 A

Q034 A

Quiz MCQ 1 A

Quiz MCQ 2 B

Quiz MCQ 3 C

Quiz MCQ 4 D

Quiz WRITTEN 5 3

Quiz WRITTEN 6 6

Quiz WRITTEN 7 12

Quiz WRITTEN 8 5

Concept 2 | Differentiation and Tangent Lines

English Student Edition • Focused formulas and guided practice

Formula opener

Gradient

$$m = f'(x_1)$$

Tangent

$$y - y_1 = m(x - x_1)$$

Normal

$$m_n m = -1$$

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Worked Solutions **Differentiation and Tangent Lines**

Q001 Written

Differentiate the four Warm Up demands: $y = x^2 - 3x + 2$, $y = (2x + 1)^3$, $V = \pi r^2 h$ with respect to (r), and find $f'(-1)$ for $f(x) = x^3 - 3x + 1$.

Solution

1. Use $(x^n)' = nx^{n-1}$. For a composite cubic, expand or use the chain rule. Treat (h) as constant in dV/dr, then substitute $x = -1$ into $3x^2 - 3$.

Final answer

$2x - 3; 6(2x + 1)^2; 2\pi rh; 0$

Q002 Written

Differentiate the six printed Try functions, the two variable-labelled functions $S = \pi r^2$ and $h = vt - \frac{1}{2}gt^2$, and evaluate $f'(2), f'(-3)$ for $f(x) = 4x^3 - 3x^2 + 2$.

Solution

1. Differentiate each term, keeping non-differentiated variables constant. For the values, first form $f'(x) = 12x^2 - 6x$, then substitute.

Final answer

$3, -6x + 8, 2x + 3, 6x^2 - 8x + 6, 6(2x - 3)^2, 4x^2 + x - 1; 2\pi r, v - gt; 36, 126$

Worked Solutions **Differentiation and Tangent Lines**

Q003 Written

Differentiate every given Exercise function $18 polynomial/product forms$, the six functions with variables indicated in brackets, and evaluate the two values in each of Exercises 3–5.

Solution

1. Bring down the exponent and subtract one. Expand products before differentiating when no chain rule is being used; then substitute the stated input.

Final answer

Apply the power rule term-by-term; for a product or power, expand first. The value sets are: Exercise 3 (33,65); Exercise 4 $-24, -3$; Exercise 5 (3,28).

Worked Solutions **Differentiation and Tangent Lines**

Q004 Written

For $f(x) = \sqrt[3]{x^5} + \frac{1}{x^2}$, find $f'(1) + f'(-1)$.

Solution

1. Write $f(x) = x^{5/3} + x^{-2}$, so $f'(x) = \frac{5}{3}x^{2/3} - 2x^{-3}$. Hence $f'(1) = -\frac{1}{3}$, $f'(-1) = \frac{11}{3}$, and the sum is $\frac{10}{3}$.

Final answer

$$\frac{10}{3}$$

Q005 Written

Find the quadratic $f(x) = ax^2 + bx + c$ satisfying $f'(0) = -2, f'(2) = 2, f(1) = 2$.

Solution

1. Since $f'(x) = 2ax + b$, $b = -2$, $4a + b = 2$, and $a + b + c = 2$. Thus $a = 1, b = -2, c = 3$.

Final answer

$f(x) = x^2 - 2x + 3$

Worked Solutions **Differentiation and Tangent Lines**

Q006 Written

Find the quadratic satisfying $f'(0) = 2, f'(1) = 4, f(2) = 6$.

Solution

1. From $2ax + b$: $b = 2$, $2a + b = 4$, and $4a + 2b + c = 6$, giving $a = 1, b = 2, c = 0$.

Final answer

$f(x) = x^2 + 2x$

Q007 Written

Find the four quadratics satisfying the four given triples of derivative and function conditions.

Solution

1. For each item set $f(x) = ax^2 + bx + c$, use $f'(x) = 2ax + b$, substitute the two derivative conditions and one function condition, then solve the three linear equations.

Final answer

$x^2 - 4x + 8; -2x^2 + 3x + 3; -5x^2 + 26x - 14; -\frac{3}{2}x^2 + 2x + \frac{3}{2}$

Worked Solutions **Differentiation and Tangent Lines**

Q008 Written

Find the quadratic (f(x)) satisfying $f'(2) + f'(4) = 0, f(3) = 0, f(1) = 12$.

Solution

1. With $f'(x) = 2ax + b$, $12a + 2b = 0$, $9a + 3b + c = 0$, and $a + b + c = 12$. Solving gives $a = 3, b = -18, c = 27$.

Final answer

$f(x) = 3x^2 - 18x + 27$

Q009 Written

Find the tangent line to $y = x^2 - 4x + 1$ at $(3, -2)$.

Solution

1. The gradient is $f'(3) = 2$. Use $y + 2 = 2(x - 3)$, hence $y = 2x - 8$.

Final answer

$y = 2x - 8$

Worked Solutions   **Differentiation and Tangent Lines**

Q010 Written

Find the tangent line for the three given functions at their stated points: $y = x^2 + 3$ at $(-2, 7)$, $y = x^2 - 3x + 2$ at ((1,0)), and $y = -x^3 + 2x + 4$ at $(-2, 8)$.

Solution

1. Differentiate, evaluate the gradient at the given x-coordinate, and use $y - y_1 = m(x - x_1)$ for each point.

Final answer

$y = -4x - 1; y = -x + 1; y = -10x - 12$

Q011 Written

Find the tangent line for all nine given Exercise functions at their given points.

Solution

1. For each item compute $m = f'(a)$, insert ((a,f(a))) into $y - f(a) = m(x - a)$, then rearrange.

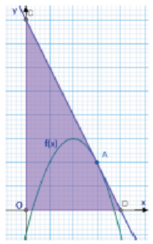
Final answer

$y = 4x - 5; y = 4x - 1; y = x - 4; y = x - 7; y = -4x + 3; y = 7x - 3; y = 9x - 16; y = -x - 2; y = x + 1$

Worked Solutions **Differentiation and Tangent Lines**

Q012 **Written**

For $f(x) = -x^2 + 4x - 1$, the tangent at x-coordinate 3 meets the axes at (D,C). Find the area of triangle (ODC).



Solution

1. The tangent is $y = -2x + 8$, so $D = (4, 0)$ and $C = (0, 8)$. Therefore area $= \frac{1}{2}(4)(8) = 16$. The ministry graph is retained beside the demand.

Final answer

16 square units

Q013 Written

Solve the two given Warm Up demands: tangents from (C(0,2)) to $y = x^2 - 3x + 6$, and tangents to $y = x^3 + 4x^2 - 3$ with gradient 3.

Solution

1. Let the point of tangency be ((a,f(a))). Equate the tangent line through the external point, or set (f'(a)) equal to the required gradient, then solve for (a).

Final answer

$y = -7x + 2, y = x + 2; \quad y = 3x + 15, y = 3x - \frac{95}{27}$

Q014 Written

Solve the two given Try demands: tangents from $C(1, -1)$ to $y = x^2 - x + 3$, and tangents to $y = x^3 + 2x - 3$ with gradient 5.

Solution

1. Use ((a,f(a))) and $m = f'(a)$. The first condition gives $a = 3, -1$; the second gives $a = 1, -1$. Substitute each point into the tangent formula.

Final answer

$y = 5x - 6, y = -3x + 2; \quad y = 5x - 5, y = 5x - 1$

Worked Solutions **Differentiation and Tangent Lines**

Q015 Written

Solve all six given Exercise demands involving external points or prescribed gradients.

Solution

1. Let the contact point be ((a,f(a))), form $y - f(a) = f'(a)(x-a)$, and impose the external point or gradient condition. Retain every real value of (a).

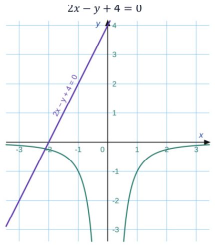
Final answer

$y = 6x - 8, y = -2x; y = x + 1, y = -7x + 25; y = -5x + 6, y = 3x - 2; y = 9x - 11, y = 9x + 21; y = -x; y = -2x + 16, y = -2x - 8$

Worked Solutions **Differentiation and Tangent Lines**

Q016 Written

For $f(x) = -1/x^2$, find the tangent line parallel to $2x - y + 4 = 0$.



Solution

1. The given line has gradient 2. Since $f'(x) = 2/x^3$, $2/a^3 = 2$ gives $a = 1$. The tangent at $(1, -1)$ is $y + 1 = 2(x - 1)$, so $y = 2x - 3$. The ministry graph is retained beside the demand.

Final answer

$y = 2x - 3$

Worked Solutions **Differentiation and Tangent Lines**

Q017 MCQ

The derivative of x^n is:

Solution

1. Bring down n and subtract 1 from the exponent.

Correct option: B

Final answer

nx^{n-1}

Q018 MCQ

The derivative of $V = \pi r^2 h$ with respect to r is:

Solution

1. Treat h as a constant.

Correct option: B

Final answer

$2\pi rh$

Worked Solutions **Differentiation and Tangent Lines**

Q019 MCQ

For $f(x) = x^3 - 3x + 1$, $f'(-1)$ is:

Solution

1. $f'(x) = 3x^2 - 3$.

Correct option: C

Final answer

0

Q020 MCQ

The challenge function can be written as:

Solution

1. Convert the cube root and reciprocal to powers.

Correct option: A

Final answer

$x^{5/3} + x^{-2}$

Worked Solutions **Differentiation and Tangent Lines**

Q021 MCQ

For $f(x) = ax^2 + bx + c$, (f'(x)) is:

Solution

1. Differentiate the quadratic.

Correct option: B

Final answer

$2ax + b$

Q022 MCQ

The condition $f'(0) = -2$ gives:

Solution

1. Substitute $x = 0$ into $2ax + b$.

Correct option: B

Final answer

$b = -2$

Worked Solutions **Differentiation and Tangent Lines**

Q023 MCQ

The warm-up quadratic is:

Solution

1. Solve the three coefficient equations.

Correct option: A

Final answer

$x^2 - 2x + 3$

Q024 MCQ

The challenge coefficient a is:

Solution

1. Use $f'(2) + f'(4) = 0$, then the two function values.

Correct option: C

Final answer

3

Worked Solutions **Differentiation and Tangent Lines**

Q025 **MCQ**

The tangent gradient at x=a is:

Solution

1. The derivative gives the instantaneous gradient.

Correct option: C

Final answer

$f'(a)$

Q026 **MCQ**

The tangent to $y = x^2 - 4x + 1$ at $(3, -2)$ is:

Solution

1. Use gradient 2 and point $3, -2$.

Correct option: B

Final answer

$y = 2x - 8$

Worked Solutions **Differentiation and Tangent Lines**

Q027 MCQ

In the triangle challenge, the x-intercept of the tangent is:

Solution

1. The tangent is $y = -2x + 8$.

Correct option: C

Final answer

4

Q028 MCQ

The area of triangle (ODC) in the challenge is:

Solution

1. Use one half base times height.

Correct option: C

Final answer

16

Worked Solutions **Differentiation and Tangent Lines**

Q029 MCQ

When the contact point is not given, set it to:

Solution

1. Use an unknown contact x-coordinate.

Correct option: B

Final answer

$(a, f(a))$

Q030 MCQ

The gradient of $2x - y + 4 = 0$ is:

Solution

1. Rearrange to $y = 2x + 4$.

Correct option: D

Final answer

2

Worked Solutions **Differentiation and Tangent Lines**

Q031 MCQ

For $f(x) = -1/x^2$, (f'(x)) is:

Solution

1. Write $f = -x^{-2}$ and differentiate.

Correct option: B

Final answer

$2/x^3$

Q032 MCQ

The parallel tangent in the challenge is:

Solution

1. The contact point is $a = 1$.

Correct option: B

Final answer

$y = 2x - 3$

Worked Solutions **Differentiation and Tangent Lines**

Quiz WRITTEN 1 Concept Quiz · Written

Differentiate $3x^3 - 2x$.

Solution

1. Apply the power rule.

Final answer

$9x^2 - 2$

Quiz WRITTEN 2 Concept Quiz · Written

Find the quadratic with $f'(0) = 1, f'(1) = 3, f(1) = 2$.

Solution

1. Use $f'(x) = 2ax + b$ and solve for (a,b,c).

Final answer

$x^2 + x$

Worked Solutions **Differentiation and Tangent Lines**

Quiz WRITTEN 3 **Concept Quiz · Written**

Find the tangent to $y = x^2$ at ((2,4)).

Solution

1. Gradient $f'(2) = 4$, then use point-slope form.

Final answer

$y = 4x - 4$

Quiz WRITTEN 4 **Concept Quiz · Written**

For $f(x) = -1/x$, find the tangent parallel to $y = x + 2$.

Solution

1. Set $f'(a) = 1$, giving $a = 1$.

Final answer

$y = x - 4$

Worked Solutions **Differentiation and Tangent Lines**

Quiz MCQ 5 **Concept Quiz · MCQ**

The derivative of a constant is:

Solution

1. A constant has zero rate of change.

Correct option: A

Final answer

0

Quiz MCQ 6 **Concept Quiz · MCQ**

For $f(x) = ax^2 + bx + c$, the coefficient equations come from:

Solution

1. There are three unknown coefficients.

Correct option: B

Final answer

two derivatives and one function value

Worked Solutions **Differentiation and Tangent Lines**

Quiz MCQ 7 **Concept Quiz · MCQ**

The tangent equation in point-slope form is:

Solution

1. Use the given point and tangent gradient.

Correct option: A

Final answer

$y - y_1 = mx - x_1$

Quiz MCQ 8 **Concept Quiz · MCQ**

A tangent parallel to $y = 3x - 1$ has gradient:

Solution

1. Parallel lines have equal gradients.

Correct option: C

Final answer

3

Answer key

Use the question number shown on each page.

Q001 $2x - 3; 6(2x+1)^2; 2\pi rh; 0$

Q002 $3, -6x+8, 2x+3, 6x^2-8x+6, 6(2x-3)^2, 4x^2+x-1; 2\pi r, v-gt; 36, 126$

Q003 Apply the power rule term-by-term; for a product or power, expand first. The value sets are: Exercise 3 (33,65); Exercise 4 −24, −3; Exercise 5 (3,28).

Q004 $\frac{10}{3}$

Q005 $f(x) = x^2 - 2x + 3$

Q006 $f(x) = x^2 + 2x$

Q007 $x^2 - 4x + 8; -2x^2 + 3x + 3; -5x^2 + 26x - 14; -\frac{3}{2}x^2 + 2x + \frac{3}{2}$

Q008 $f(x) = 3x^2 - 18x + 27$

Q009 $y = 2x - 8$

Q010 $y = -4x - 1; y = -x + 1; y = -10x - 12$

Q011 $y = 4x - 5; y = 4x - 1; y = x - 4; y = x - 7; y = -4x + 3; y = 7x - 3; y = 9x - 16; y = -x - 2; y = x + \ldots$

Q012 16 square units

Q013 $y = -7x + 2, y = x + 2; \quad y = 3x + 15, y = 3x - \frac{95}{27}$

Q014 $y = 5x - 6, y = -3x + 2; \quad y = 5x - 5, y = 5x - 1$

Q015 $y = 6x - 8, y = -2x; y = x + 1, y = -7x + 25; y = -5x + 6, y = 3x - 2; y = 9x - 11, y = 9x + 21; y = -x; y = -2x + 16, y = -2x - 8$

Q016 $y = 2x - 3$

Q017 B

Q018 B

Q019 C

Q020 A

Q021 B

Q022 B

Q023 A

Q024 C

Q025 C

Q026 B

Q027 C

Q028 C

Q029 B

Q030 D

Q031 B

Q032 B

Quiz WRITTEN 1 $9x^2 - 2$

Quiz WRITTEN 2 $x^2 + x$

Quiz WRITTEN 3 $y = 4x - 4$

Quiz WRITTEN 4 $y = x - 4$

Quiz MCQ 5 A

Quiz MCQ 6 B

Quiz MCQ 7 A

Quiz MCQ 8 C

Concept 3 | Applications of Differentiation: Polynomial Functions

English Student Edition • Focused formulas and guided practice

Formula opener

Rate

$$\frac{dy}{dx}$$

Stationary

$$f'(x) = 0$$

Second derivative

$$f''(x) > 0 \Rightarrow \text{minimum}$$

Eng Eslam Ahmed | 01120009622

Worked Solutions **Applications of Differentiation: Polynomial Functions**

Q001 Written

Find the extreme values and sketch the graphs of the three cubic functions printed in the Warm Up.

Solution

1. Differentiate, solve $f'(x) = 0$, make a sign table, and substitute each stationary x-value into $f(x)$.

Final answer

Use the derivative sign table: for $x^3 - 6x^2 + 16$, local maximum 16 at $x = 0$ and local minimum -16 at $x = 4$; classify the other two from their stationary points.

Q002 Written

Find the extreme values and sketch the three given Try cubics.

Solution

1. Record the sign of f' from left to right and label each change $+ \to -$ as a local maximum and $- \to +$ as a local minimum.

Final answer

Apply the same derivative sign-table test; a stationary point where the sign of f' does not change is not an extreme value.

Worked Solutions **Applications of Differentiation: Polynomial Functions**

Q003 **Written**

Find the extreme values and draw the graphs for all nine given cubic functions in Exercise.

Solution

1. Use exact stationary coordinates and keep the graph's increasing/decreasing intervals consistent with the sign table.

Final answer

Differentiate each polynomial, solve the quadratic derivative, use the sign table, and evaluate f at every genuine stationary point.

Q004 **Written**

For $P(x) = x^3 - 3ax^2 + (3a^2 - 3)x + 4$, $a > 0$, find the flow rates giving the maximum and minimum peak efficiency.

Solution

1. $P'(x) = 3[(x-a)^2 - 1]$, so the critical points are $a - 1$ and $a + 1$. The derivative changes $+ \to -$ then $- \to +$.

Final answer

Local maximum at $x = a - 1$; local minimum at $x = a + 1$.

Worked Solutions **Applications of Differentiation: Polynomial Functions**

Q005 Written

On $[-3, 4]$, find the maximum and minimum values of $y = -x^3 + 12x + 5$.

Solution

1. Compare $f(-3)$, $f(4)$ with the stationary values $f(-2) = -11$ and $f(2) = 21$.

Final answer

Maximum 21 at $x = 2$; minimum -11 at $x = -2$.

Q006 Written

Find the maximum and minimum values of the two given Try cubics on their stated closed intervals.

Solution

1. A local extreme is not automatically absolute: compare it with both domain endpoints.

Final answer

Evaluate both endpoints and every stationary point inside the interval; the greatest value is the absolute maximum and the least is the absolute minimum.

Worked Solutions **Applications of Differentiation: Polynomial Functions**

Q007 Written

Find the maximum and minimum values of all six given cubics on their given closed intervals.

Solution

1. State the extreme value together with its x-coordinate and retain only points in the specified domain.

Final answer

For each function, compare the endpoint values with the values at all critical x-values lying in the interval.

Q008 Written

For $f(x) = x^3 - 6x$ on $[-3, 3]$, find the absolute deepest dive and highest ascent.

Solution

1. Critical points are $x = \pm\sqrt{2}$, with values $\mp 4\sqrt{2}$; compare these with $f(-3) = -9$ and $f(3) = 9$.

Final answer

Absolute minimum -9 at $x = -3$; absolute maximum 9 at $x = 3$.

Worked Solutions **Applications of Differentiation: Polynomial Functions**

Q009 Written

If $f(x) = -x^3 + ax + b$ has a local maximum of 9 at $x = 2$, find a, b, and the local minimum.

Solution

1. Use $f'(2) = 0$ to get $a = 12$, then $f(2) = 9$ to get $b = -7$. The other stationary point is $x = -2$.

Final answer

$a = 12, b = -7$; local minimum -23 at $x = -2$.

Q010 Written

Solve the two given Try problems: determine the coefficients from a local minimum/maximum and another stationary point.

Solution

1. Differentiate first, use each stated extreme point, then evaluate the remaining extreme value from the resulting cubic.

Final answer

Use $f'(a) = 0, f(a) = M$, and the second stationary condition to form three simultaneous equations in the coefficients.

Worked Solutions **Applications of Differentiation: Polynomial Functions**

Q011 Written

Solve all six given Exercise problems asking for cubic coefficients from extrema.

Solution

1. Do not skip the condition $f'(a) = 0$: it identifies the stationary point before the function-value equation is used.

Final answer

Each set is determined by two derivative roots/conditions and one extreme-value condition; solve the resulting simultaneous equations.

Q012 Written

For $f(x) = x^3 + ax^2 + bx + c$, the graph has a local peak 178 at $x = -2$ and a local trough at $x = 4$. Find a, b, c.

Solution

1. The derivative roots are $-2, 4$, so $f'(x) = 3(x+2)(x-4) = 3x^2 - 6x - 24$. Thus $a = -3, b = -24$. Substitution of $f(-2) = 178$ gives $c = 150$.

Final answer

$a = -3,\ b = -24,\ c = 150.$

Worked Solutions **Applications of Differentiation: Polynomial Functions**

Q013 Written

If $f(x) = x^3 - 3kx^2 + kx + 2$ has no extreme values, find the range of k.

Solution

1. The derivative is $f'(x) = 3x^2 - 6kx + k$. No two distinct stationary roots means $D = 36k^2 - 12k \leq 0$, hence $0 \leq k \leq \frac{1}{3}$.

Final answer

$0 \leq k \leq \frac{1}{3}$.

Q014 Written

Determine the parameter ranges in the given Try problems for a cubic to have no extreme values or to have both extremes.

Solution

1. Differentiate, calculate the quadratic discriminant, and solve the resulting parameter inequality exactly.

Final answer

Use $D \leq 0$ for no extremes and $D > 0$ for two distinct stationary points.

Worked Solutions **Applications of Differentiation: Polynomial Functions**

Q015 **Written**

Determine the parameter ranges for all six given Exercise cubics according to whether they have extreme values.

Solution

1. Remember: $D > 0$ gives two distinct real stationary roots; $D \leq 0$ gives no pair of distinct extremes.

Final answer

Apply the derivative-discriminant criterion separately to each printed cubic.

Q016 **Written**

For $f(x) = x^3 + 3kx^2 + 12kx + 10$, find the exact k-range that guarantees both a local maximum and minimum.

Solution

1. $f'(x) = 3x^2 + 6kx + 12k$ has two distinct real roots when $36k^2 - 144k > 0$, i.e. $k(k-4) > 0$.

Final answer

$k < 0 or k > 4$.

Worked Solutions **Applications of Differentiation: Polynomial Functions**

Q017 **Written**

If $f(x) = x^3 - 3kx^2 + (k^2 + 5)x + 3$ is always monotonically increasing, find the range of k.

Solution

1. For a positive leading coefficient, monotonic increase requires the derivative quadratic to have no two distinct real roots: $D = 24k^2 - 60 \leq 0$.

Final answer

$-\frac{\sqrt{10}}{2} \leq k \leq \frac{\sqrt{10}}{2}$.

Q018 **Written**

Find the parameter ranges in the two given Try problems for the cubic to be monotonically increasing.

Solution

1. Monotonicity is checked from $f'(x) \geq 0$ for every real x; use the discriminant condition for the quadratic derivative.

Final answer

Set the derivative discriminant $D \leq 0$, with the derivative leading coefficient positive.

Worked Solutions **Applications of Differentiation: Polynomial Functions**

Q019 Written

Find the parameter ranges in all six given Exercise problems for monotonic increase.

Solution

1. Complete the discriminant test and include equality when the derivative has a repeated zero.

Final answer

Differentiate each cubic and impose that its quadratic derivative never becomes negative.

Q020 Written

For $f(x) = \frac{1}{3}x^3 - kx^2 + (k+6)x + 10$, find the k-range for monotonic increase on $x \in [1, \infty)$.

Solution

1. $f'(x) = x^2 - 2kx + k + 6$. If $k \leq 1$, its minimum on $[1, \infty)$ is $7 - k \geq 0$; if $k > 1$, the vertex condition gives $k + 6 - k^2 \geq 0$, hence $k \leq 3$.

Final answer

$k \leq 3$.

Worked Solutions **Applications of Differentiation: Polynomial Functions**

Q021 **Written**

Find the number of distinct real roots of $x^3 - 6x^2 + 9x - 3 = 0$.

Solution

1. The stationary values are $f(1) = 1 > 0$ and $f(3) = -3 < 0$, so the cubic crosses the x-axis three times.

Final answer

Three distinct real roots.

Q022 **Written**

Find the number of distinct real roots for the three given Try equations.

Solution

1. Draw $y = f(x)$, find its critical points, and compare the local maximum/minimum with zero.

Final answer

Use the cubic graph: count x-axis intersections from the stationary points and end behavior.

Worked Solutions **Applications of Differentiation: Polynomial Functions**

Q023 **Written**

Find the number of distinct real roots for all nine given cubic equations.

Solution

1. Differentiate, locate the turning points, and use their signs relative to zero to decide between one, two, or three distinct roots.

Final answer

Each answer is obtained from the number of x-axis intersections of its corresponding cubic graph.

Q024 **Written**

Find the number of distinct real roots of $-h^3 + 12h^2 - 36h + 40 = 0$.

Solution

1. The derivative is $-3(h-2)(h-6)$; both stationary values are positive, so the graph crosses the axis only once.

Final answer

One distinct real root.

Worked Solutions **Applications of Differentiation: Polynomial Functions**

Q025 Written

If $-x^3 + 3x^2 + 9x + a = 0$ has three distinct real roots, find the range of a.

Solution

1. Rewrite as $x^3 - 3x^2 - 9x = a$. The local values are 5 and -27; a horizontal line intersects the cubic three times strictly between them.

Final answer

$-27 < a < 5$.

Q026 Written

Solve the two given Try parameter problems for three or two distinct real roots.

Solution

1. Strict inequalities give three roots; equality at a stationary value produces two distinct roots and must be handled separately.

Final answer

Rewrite each equation as $g(x) = a$, then compare the parameter line with the local maximum and minimum of g.

Worked Solutions **Applications of Differentiation: Polynomial Functions**

Q027 Written

Solve all four given Exercise parameter problems for the required number of distinct roots.

Solution

1. Keep the endpoint cases: tangency at an extreme value counts as two distinct roots, not three.

Final answer

Use the local extrema of the parameter-free cubic and count intersections with the horizontal parameter line.

Q028 Written

For $f(x) = x^3 - 6x^2 + 9x - 2a^2 + 7a$, find all real a such that $f(x) = 2$ has exactly three distinct real roots.

Solution

1. Set $g(x) = x^3 - 6x^2 + 9x$, whose extrema are 0 and 4. The horizontal level is $2a^2 - 7a + 2$; require $0 < 2a^2 - 7a + 2 < 4$.

Final answer

$\frac{7-\sqrt{65}}{4} < a < \frac{7-\sqrt{33}}{4} or \frac{7+\sqrt{33}}{4} < a < \frac{7+\sqrt{65}}{4}$.

Worked Solutions **Applications of Differentiation: Polynomial Functions**

Q029 Written

Prove the given inequalities using an auxiliary cubic and a derivative sign table.

Solution

1. Differentiate the auxiliary function, make the sign table on the domain, and substitute the minimum point or endpoint.

Final answer

Set $f(x) = \text{LHS} - \text{RHS}$, prove $f(x) \geq 0$ on the stated domain, and identify its minimum.

Q030 Written

Prove all three given cubic inequalities on their stated domains.

Solution

1. The proof is complete only after the derivative sign table and the minimum value are stated.

Final answer

Move the right-hand side to the left, then show the resulting auxiliary cubic has non-negative minimum on the domain.

Worked Solutions **Applications of Differentiation: Polynomial Functions**

Q031 Written

Find the largest k such that $x^3 - 3x^2 - 3x + k + 3 \le 0$ for every $x \in [-1, 3]$.

Solution

1. The maximum of $x^3 - 3x^2 - 3x + 3$ on the interval is $4\sqrt{2} - 2$ at $x = 1 - \sqrt{2}$. Hence $k + 4\sqrt{2} - 2 \le 0$.

Final answer

$k \le 2 - 4\sqrt{2}$; the largest value is $2 - 4\sqrt{2}$.

Q032 Written

Find the extreme values and sketch the two given quartic functions.

Solution

1. Differentiate the quartic, use a sign table for the cubic derivative, and substitute the critical values.

Final answer

For $3x^4 + 4x^3 - 12x^2$: local maximum 0 at $x = 0$, local minima -32 at $x = -2$ and -5 at $x = 1$. For $-x^4 - 4x^3 - 8$: maximum 19 at $x = -3$; $x = 0$ is stationary but not an extreme.

Worked Solutions **Applications of Differentiation: Polynomial Functions**

Q033 **Written**

Find the extreme values and graphs for all six given quartic functions.

Solution

1. Repeated derivative roots may be stationary without being extrema; check the sign change before labelling a point.

Final answer

Differentiate each quartic, construct the derivative sign table, and evaluate each genuine turning point.

Q034 **Written**

For $f(x) = x^4 - 4x^3 + k$, find the exact k-range for exactly two distinct real roots.

Solution

1. The global minimum of $x^4 - 4x^3 + k$ is $k - 27$ at $x = 3$. Exactly two crossings require this minimum to be negative.

Final answer

$k < 27$.

Worked Solutions **Applications of Differentiation: Polynomial Functions**

Q035 MCQ

At a local maximum, the sign of f' changes:

Solution

1. Use the first derivative sign change.

Correct option: B

Final answer

$+ to -$

Q036 MCQ

For $P'(x) = 3[(x-a)^2 - 1]$, the local maximum occurs at:

Solution

1. The derivative changes + to - at $a - 1$.

Correct option: A

Final answer

$a - 1$

Worked Solutions **Applications of Differentiation: Polynomial Functions**

Q037 *MCQ*

Absolute extrema on a closed interval require comparing:

Solution

1. Closed intervals include both.

Correct option: C

Final answer

stationary points and endpoints

Q038 *MCQ*

The absolute maximum of $x^3 - 6x$ on $[-3, 3]$ is:

Solution

1. Compare the endpoints with the critical values.

Correct option: B

Final answer

9

Worked Solutions **Applications of Differentiation: Polynomial Functions**

Q039 MCQ

If $f(x) = -x^3 + ax + b$ has a stationary point at 2, then a is:

Solution

1. Set $f'(2) = -12 + a = 0$.

Correct option: C

Final answer

12

Q040 MCQ

The derivative roots in the challenge are:

Solution

1. Use the two turning-point x-values.

Correct option: B

Final answer

-2 and 4

Worked Solutions **Applications of Differentiation: Polynomial Functions**

Q041 MCQ

No two distinct extrema for a cubic means the derivative discriminant is:

Solution

1. A repeated root or no real roots gives no pair.

Correct option: C

Final answer

$D <= 0$

Q042 MCQ

The challenge range for both extrema is:

Solution

1. Require $36k^2 - 144k > 0$.

Correct option: B

Final answer

$k < 0 or k > 4$

Worked Solutions **Applications of Differentiation: Polynomial Functions**

Q043 MCQ

For a positive-leading cubic to be monotone increasing, its derivative must be:

Solution

1. Use $f'(x) \geq 0$.

Correct option: B

Final answer

non-negative everywhere

Q044 MCQ

The challenge domain in 7-13 is:

Solution

1. Use the restricted domain in the vertex test.

Correct option: B

Final answer

[1,infinity)

Worked Solutions **Applications of Differentiation: Polynomial Functions**

Q045 MCQ

Three distinct roots occur when the x-axis lies:

Solution

1. Then the graph crosses the axis three times.

Correct option: B

Final answer

between the extrema

Q046 MCQ

The 7-14 challenge has how many distinct real roots?

Solution

1. Both turning values are above zero.

Correct option: B

Final answer

1

Worked Solutions **Applications of Differentiation: Polynomial Functions**

Q047 MCQ

Three intersections with $y = a$ require the level to lie:

Solution

1. Tangency is an endpoint case.

Correct option: A

Final answer

strictly between the extrema

Q048 MCQ

The 7-15 warm-up range is:

Solution

1. Use the two local values.

Correct option: A

Final answer

$-27 < a < 5$

Worked Solutions **Applications of Differentiation: Polynomial Functions**

Q049 MCQ

To prove LHS $\geq$ RHS, define:

Solution

1. Then prove $f \geq 0$.

Correct option: B

Final answer

LHS-RHS

Q050 MCQ

The largest k in the challenge is:

Solution

1. Bound k by the maximum of the k-free part.

Correct option: A

Final answer

2-4sqrt(2)

Worked Solutions **Applications of Differentiation: Polynomial Functions**

Q051 MCQ

A repeated root of a derivative may be:

Solution

1. Check the sign change.

Correct option: C

Final answer

stationary but not an extreme

Q052 MCQ

Exactly two distinct roots for $x^4 - 4x^3 + k = 0$ require:

Solution

1. The global minimum must be below zero.

Correct option: B

Final answer

$k < 27$

Worked Solutions **Applications of Differentiation: Polynomial Functions**

Q053 MCQ

A local minimum is identified by the change:

Solution

1. A negative derivative followed by a positive derivative gives a local minimum.

Correct option: B

Final answer

$-to+$

Q054 MCQ

For $y = x^3 - 6x^2 + 16$, the critical x-values are:

Solution

1. Solve $3x^2 - 12x = 0$.

Correct option: A

Final answer

0 and 4

Worked Solutions **Applications of Differentiation: Polynomial Functions**

Q055 MCQ

The derivative sign table is used to determine:

Solution

1. Signs of (f') control monotonicity.

Correct option: B

Final answer

increase/decrease and extrema

Q056 MCQ

On a closed interval, an endpoint can be:

Solution

1. Compare both endpoints with stationary values.

Correct option: B

Final answer

an absolute extreme

Worked Solutions **Applications of Differentiation: Polynomial Functions**

Q057 MCQ

For $y = -x^3 + 12x + 5$, the stationary x-values are:

Solution

1. Solve $-3x^2 + 12 = 0$.

Correct option: A

Final answer

-2 and 2

Q058 MCQ

The absolute minimum of $x^3 - 6x$ on $[-3, 3]$ is at:

Solution

1. Compare all candidates; $f(-3) = -9$.

Correct option: A

Final answer

$x = -3$

Worked Solutions **Applications of Differentiation: Polynomial Functions**

Q059 MCQ

To determine a cubic from an extreme value, use both $f(a) = M$ and:

Solution

1. An extreme point is stationary.

Correct option: B

Final answer

$f'(a) = 0$

Q060 MCQ

In the warm-up $f(x) = -x^3 + 12x - 7$, the local minimum occurs at:

Solution

1. The derivative changes from negative to positive at -2.

Correct option: A

Final answer

$x = -2$

Worked Solutions **Applications of Differentiation: Polynomial Functions**

Q061 *MCQ*

For $f(x) = x^3 - 3kx^2 + kx + 2$, no two distinct stationary roots requires:

Solution

1. Use $D = 12k(3k - 1) \leq 0$.

Correct option: A

Final answer

$0 <= k <= 1/3$

Q062 *MCQ*

Two distinct stationary points occur when the derivative discriminant is:

Solution

1. A positive discriminant gives two distinct real derivative roots.

Correct option: C

Final answer

$D > 0$

Worked Solutions **Applications of Differentiation: Polynomial Functions**

Q063 MCQ

For a positive-leading quadratic derivative, $D \leq 0$ implies it is:

Solution

1. The quadratic has no two distinct real zeros.

Correct option: A

Final answer

non-negative for all x

Q064 MCQ

In the 7-13 challenge, monotonicity is required on:

Solution

1. Use the operational interval.

Correct option: A

Final answer

[1,infinity)

Worked Solutions **Applications of Differentiation: Polynomial Functions**

Q065 MCQ

A cubic with both stationary values above the x-axis has:

Solution

1. The graph crosses the x-axis once.

Correct option: B

Final answer

one distinct real root

Q066 MCQ

To count cubic roots, compare the extrema with:

Solution

1. Roots are x-axis intersections.

Correct option: B

Final answer

the x-axis

Worked Solutions **Applications of Differentiation: Polynomial Functions**

Q067 MCQ

At an extremal parameter level, a cubic and a horizontal line have:

Solution

1. One contact is tangent and the other crossing remains.

Correct option: B

Final answer

two distinct intersections

Q068 MCQ

For $x^4 - 4x^3 + k$, the global minimum occurs at:

Solution

1. The derivative is $4x^2(x-3)$.

Correct option: C

Final answer

$x = 3$

Worked Solutions **Applications of Differentiation: Polynomial Functions**

Quiz WRITTEN 1 **Concept Quiz · Written**

Find the local maximum and minimum of $x^3 - 3x^2 + 1$.

Solution

1. Use $f'(x) = 3x(x-2)$ and a sign table.

Final answer

Maximum 1 at $x = 0$; minimum -3 at $x = 2$

Quiz WRITTEN 2 **Concept Quiz · Written**

Find a, b if $f(x) = -x^3 + ax + b$ has a local maximum 4 at x=1.

Solution

1. Use $f'(1) = 0$ and $f(1) = 4$.

Final answer

$a = 3,\ b = 2$

Worked Solutions **Applications of Differentiation: Polynomial Functions**

Quiz WRITTEN 3 **Concept Quiz · Written**

Find the k-range for $x^3 - 3kx^2 + kx$ to have no two distinct stationary points.

Solution

1. Set the derivative discriminant ≤ 0.

Final answer

$-\frac{\sqrt{10}}{2} \leq k \leq \frac{\sqrt{10}}{2}$

Quiz WRITTEN 4 **Concept Quiz · Written**

Find the largest k such that $x^3 - 3x^2 - 3x + k + 3 \leq 0$ on $[-1, 3]$.

Solution

1. Use the maximum of the k-free cubic on the interval.

Final answer

$2 - 4\sqrt{2}$

Worked Solutions **Applications of Differentiation: Polynomial Functions**

Quiz MCQ 5 **Concept Quiz · MCQ**

At a local minimum, f' changes:

Solution

1. Use the first derivative sign change.

Correct option: B

Final answer

$-to+$

Quiz MCQ 6 **Concept Quiz · MCQ**

A cubic with a positive-leading derivative and $D < 0$ is:

Solution

1. The derivative quadratic stays positive.

Correct option: A

Final answer

always increasing

Worked Solutions **Applications of Differentiation: Polynomial Functions**

Quiz MCQ 7 **Concept Quiz · MCQ**

The 7-15 challenge requires the level to satisfy:

Solution

1. Three distinct intersections lie strictly between extrema.

Correct option: B

Final answer

$0 < t < 4$

Quiz MCQ 8 **Concept Quiz · MCQ**

The global minimum of $x^4 - 4x^3 + k$ is:

Solution

1. Evaluate at the quartic's global minimum $x = 3$.

Correct option: A

Final answer

$k - 27$

Answer key

Use the question number shown on each page.

Q001 Use the derivative sign table: for x^3-6x^2+16, local maximum 16 at $x=0$ and local minimum -16 at $x=4$; classify the other two from their stationary points.

Q002 Apply the same derivative sign-table test; a stationary point where the sign of f' does not change is not an extreme value.

Q003 Differentiate each polynomial, solve the quadratic derivative, use the sign table, and evaluate f at every genuine stationary point.

Q004 Local maximum at $x=a-1$; local minimum at $x=a+1$.

Q005 Maximum 21 at $x=2$; minimum -11 at $x=-2$.

Q006 Evaluate both endpoints and every stationary point inside the interval; the greatest value is the absolute maximum and the least is the absolute minimum.

Q007 For each function, compare the endpoint values with the values at all critical x-values lying in the interval.

Q008 Absolute minimum -9 at $x=-3$; absolute maximum 9 at $x=3$.

Q009 $a=12, b=-7$; local minimum -23 at $x=-2$.

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Q014 Use $D\le 0$ for no extremes and $D>0$ for two distinct stationary points.

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Q017 $-\frac{\sqrt{10}}{2}\le k\le \frac{\sqrt{10}}{2}.$

Q018 Set the derivative discriminant $D\le 0$, with the derivative leading coefficient positive.

Q019 Differentiate each cubic and impose that its quadratic derivative never becomes negative.

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Q021 Three distinct real roots.

Q022 Use the cubic graph: count x-axis intersections from the stationary points and end behavior.

Q023 Each answer is obtained from the number of x-axis intersections of its corresponding cubic graph.

Q024 One distinct real root.

Q025 $-27<a<5.$

Q026 Rewrite each equation as $g(x)=a$, then compare the parameter line with the local maximum and minimum of g.

Q027 Use the local extrema of the parameter-free cubic and count intersections with the horizontal parameter line.

Q028 $\frac{7-\sqrt{65}}{4}<a<\frac{7-\sqrt{33}}{4} or \frac{7+\sqrt{33}}{4}<a<\frac{7+\sqrt{65}}{4}.$

Q029 Set $f(x)=\mathrm{LHS}-\mathrm{RHS}$, prove $f(x)\ge 0$ on the stated domain, and identify its minimum.

Q030 Move the right-hand side to the left, then show the resulting auxiliary cubic has non-negative minimum on the domain.

Q031 $k\le 2-4\sqrt{2}$; the largest value is $2-4\sqrt{2}$.

Q032 For $3x^4+4x^3-12x^2$: local maximum 0 at $x=0$, local minima -32 at $x=-2$ and -5 at $x=1$. For $-x^4-4x^3-8$: maximum 19 at $x=-3$; $x=0$ is stationary but not an extreme.

Q033 Differentiate each quartic, construct the derivative sign table, and evaluate each genuine turning point.

Q034 $k<27.$

Q035 B

Q036 A

Q037 C

Q038 B

Q039 C

Q040 B

Q041 C

Q042 B

Q043 B

Q044 B

Q045 B

Q046 B

Q047 A

Q048 A

Q049 B

Q050 A

Q051 C

Q052 B

Q053 B

Q054 A

Q055 B

Q056 B

Q057 A

Q058 A

Q059 B

Q060 A

Q061 A

Q062 C

Q063 A

Q064 A

Q065 B

Q066 B

Q067 B

Q068 C

Quiz WRITTEN 1 Maximum 1 at $x = 0$; minimum -3 at $x = 2$

Quiz WRITTEN 2 $a = 3,\ b = 2$

Quiz WRITTEN 3 $-\frac{\sqrt{10}}{2} \le k \le \frac{\sqrt{10}}{2}$

Quiz WRITTEN 4 $2 - 4\sqrt{2}$

Quiz MCQ 5 B

Quiz MCQ 6 A

Quiz MCQ 7 B

Quiz MCQ 8 A

Concept 4 | Integration: Definite Integrals and Applications

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Formula opener

Antiderivative

$$\int x^n dx = \frac{x^{n+1}}{n+1} + C$$

Definite

$$\int_a^b f(x)dx = F(b) - F(a)$$

Area

$$A = \int_a^b |f(x)|dx$$

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Worked Solutions **Integration: Definite Integrals and Applications**

Q001 **Written**

Find the two indefinite integrals in the Warm Up.

Solution

1. Use $\int x^n \, dx = \frac{x^{n+1}}{n+1} + C$, then differentiate your result to check.

Final answer

Integrate term by term and add C to each primitive.

Q002 **Written**

Find the six indefinite integrals in the given Try section, expanding a squared bracket first when needed.

Solution

1. For $(x-1)^2$, write $x^2 - 2x + 1$ before applying the power rule.

Final answer

Expand products or powers, collect like powers, then integrate every term.

Worked Solutions **Integration: Definite Integrals and Applications**

Q003 **Written**

Solve the first ten given indefinite-integral exercises (constants, linear and quadratic polynomials).

Solution

1. A constant integrates to a constant times x; combine the terms only after integration.

Final answer

Apply linearity and the power rule; retain the constant of integration.

Q004 **Written**

Solve the remaining eight given integrals, including products and squared brackets.

Solution

1. Do not integrate a product factor-by-factor; expand first, then use the power rule.

Final answer

Expand each product or square, simplify, and integrate term by term.

Worked Solutions **Integration: Definite Integrals and Applications**

Q005 **Written**

Find $F(x)$ when $F'(x) = 3x^2 + 8x - 1$ and $F(1) = 7$.

Solution

1. Integrate $F'(x)$, substitute $x = 1$, and solve the resulting equation for C.

Final answer

$F(x) = x^3 + 4x^2 - x + 3.$

Q006 **Written**

Find the two given Try functions from their derivatives and one given value of F.

Solution

1. The initial value fixes the otherwise undetermined constant of integration.

Final answer

Integrate the derivative and use the given point to determine C.

Worked Solutions **Integration: Definite Integrals and Applications**

Q007 Written

Find $F(x)$ for all six given Exercise conditions.

Solution

1. Keep the derivative and the initial condition as two separate equations before solving for C.

Final answer

For each item, integrate $F'(x)$, then impose the stated value of F.

Q008 Written

Find the curve $y = f(x)$ through $(3, 2)$ whose tangent gradient is $3x^2 - 2x + 1$.

Solution

1. The gradient is $f'(x)$; integrate it and use $f(3) = 2$ to determine C.

Final answer

$f(x) = x^3 - x^2 + x - 19.$

Worked Solutions **Integration: Definite Integrals and Applications**

Q009 Written

Solve the given Try tangent-gradient problem through $(1, 1)$.

Solution

1. Replace the gradient by $f'(x)$, integrate, then substitute the known coordinate.

Final answer

Integrate the given gradient and use the point on the curve to determine the constant.

Q010 Written

Solve the three given tangent-gradient exercises, finding each function from its point and gradient.

Solution

1. A point (x_0, y_0) gives $f(x_0) = y_0$, which fixes C.

Final answer

Integrate the polynomial gradient and apply the given point condition in each case.

Worked Solutions **Integration: Definite Integrals and Applications**

Q011 Written

Evaluate the three definite integrals in the Warm Up using the Fundamental Theorem of Calculus.

Solution

1. Write $[F(x)]_a^b = F(b) - F(a)$ before substituting the limits.

Final answer

Find an antiderivative F and calculate $F(b) - F(a)$; equal limits give zero.

Q012 Written

Evaluate the six given Try definite integrals, simplifying the integrand before applying the limits.

Solution

1. Linearity allows sums and differences of integrals to be combined before integration.

Final answer

Expand products, integrate, and evaluate the antiderivative at the upper and lower limits.

Worked Solutions **Integration: Definite Integrals and Applications**

Q013 **Written**

Evaluate the first twelve given Exercise definite integrals, including reversed and equal limits.

Solution

1. Check the order of the limits before evaluating the antiderivative.

Final answer

Use the Fundamental Theorem; a reversed interval changes the sign and equal limits give 0.

Q014 **Written**

Evaluate the remaining twelve given Exercise integrals, including sums, differences and products.

Solution

1. Combine only integrals with the same limits; otherwise split or evaluate separately.

Final answer

Use linearity, expand products, then apply the endpoint-substitution rule.

Worked Solutions **Integration: Definite Integrals and Applications**

Q015 **Written**

Use the four properties of definite integrals to simplify the three Warm Up expressions.

Solution

1. Choose the property that makes the integrand or limits match a known form.

Final answer

Scale constants, reverse limits, split intervals, or use the factored-interval identity as appropriate.

Q016 **Written**

Solve the four given Try definite-integral transformations.

Solution

1. For $(x-a)(x-b)$, first make the coefficient of x equal to 1.

Final answer

Swap limits when necessary, join adjacent intervals, and normalize a linear factor before using the identity.

Worked Solutions **Integration: Definite Integrals and Applications**

Q017 Written

Solve the first six given Exercise questions using interval splitting and reversal.

Solution

1. A minus sign before an integral can be absorbed by reversing its limits.

Final answer

Rewrite each expression until adjacent intervals have matching integrands and compatible limits.

Q018 Written

Solve the final six given Exercise questions using the factorized definite-integral identity.

Solution

1. Identify (a,b) from the two linear factors and account for every constant multiplier.

Final answer

Scale to $\int_a^b (x-a)(x-b)\,dx = -\frac{(b-a)^3}{6}$.

Worked Solutions **Integration: Definite Integrals and Applications**

Q019 Written

Find $f(x)$ from the given equation $f(x) = x^2 + \int_0^1 f(t)\,dt$.

Solution

1. Replace the definite integral by a constant, substitute the resulting $f(t)$ back into it, and solve.

Final answer

Let $A = \int_0{}^1 f(t)\,dt$; then $A = frac13 + A$, so $A = -frac13$ and $f(x) = x^2 - frac13$.

Q020 Written

Solve the two given Try integral equations for $f(x)$.

Solution

1. The unknown integral is a number independent of x; determine it before writing the final function.

Final answer

Introduce a constant for the definite integral, evaluate it using the proposed $f(t)$, and solve the linear equation.

Worked Solutions **Integration: Definite Integrals and Applications**

Q021 Written

Find $f(x)$ for all five given integral equations.

Solution

1. Check the final function by integrating it over the stated interval and substituting back.

Final answer

Let the definite integral be A, substitute the expression for $f(t)$, solve for A, then recover $f(x)$.

Q022 Written

Differentiate $\int_3^x (2t^2 + 3t - 5)\, dt$ with respect to x.

Solution

1. By the Fundamental Theorem, $\frac{d}{dx}\int_a^x f(t)\, dt = f(x)$.

Final answer

$2x^2 + 3x - 5$.

Q023 Written

Solve the two given Try problems involving differentiation of a variable-limit integral and an unknown lower limit.

Solution

1. A constant lower limit disappears under differentiation; it is recovered from the original equation.

Final answer

Replace t by x when differentiating, then use equal limits or the given condition to determine a.

Q024 Written

Differentiate the three given expressions with respect to x.

Solution

1. After differentiation, replace t with the variable upper limit x.

Final answer

Apply the variable-limit Fundamental Theorem directly to each integrand.

Worked Solutions **Integration: Definite Integrals and Applications**

Q025 Written

Find $f(x)$ and a for the three given equations with a variable upper limit.

Solution

1. Use $0 = \int_a^a f(t)\,dt$ in the original equation after finding $f(x)$.

Final answer

Differentiate to get $f(x)$, then set the integral to zero at $x = a$ to solve for a.

Q026 MCQ

The constant of integration is required because:

Solution

1. Differentiation removes additive constants.

Correct option: B

Final answer

primitives differ by a constant

Worked Solutions **Integration: Definite Integrals and Applications**

Q027 MCQ

The integral of x^n for $n \neq -1$ is:

Solution

1. Use the power rule for integration.

Correct option: B

Final answer

$x^(n+1)/(n+1)+C$

Q028 MCQ

Before integrating $(x-1)^2$, first:

Solution

1. Write $x^2 - 2x + 1$.

Correct option: B

Final answer

expand it

Worked Solutions **Integration: Definite Integrals and Applications**

Q029 MCQ

The integral of $3x^2 - 2x + 1$ is:

Solution

1. Integrate term by term.

Correct option: A

Final answer

$x^3 - x^2 + x + C$

Q030 MCQ

To find $F(x)$ from $F'(x)$, you should first:

Solution

1. Integration reverses differentiation.

Correct option: B

Final answer

integrate F'x

Worked Solutions **Integration: Definite Integrals and Applications**

Q031 MCQ

The condition $F(1) = 7$ determines:

Solution

1. Use the given function value after integrating.

Correct option: C

Final answer

C

Q032 MCQ

If $F'(x) = 6x - 2$ and $F(0) = 1$, then $F(x)$ is:

Solution

1. Integrate and use $F(0) = 1$.

Correct option: A

Final answer

$3x^2 - 2x + 1$

Worked Solutions **Integration: Definite Integrals and Applications**

Q033 MCQ

The gradient of the tangent at (x, y) is:

Solution

1. By definition, the derivative is the tangent gradient.

Correct option: B

Final answer

$f'x$

Q034 MCQ

A point (x_0, y_0) on $y = f(x)$ gives:

Solution

1. Use the point as an initial condition.

Correct option: B

Final answer

$f(x0) = y0$

Worked Solutions **Integration: Definite Integrals and Applications**

Q035 MCQ

If $f'(x) = 3x^2 - 2x + 1$, an antiderivative is:

Solution

1. Integrate each term.

Correct option: B

Final answer

$x^3 - x^2 + x + C$

Q036 MCQ

The Fundamental Theorem gives $\int_a^b f(x)dx =$:

Solution

1. Evaluate an antiderivative at the upper minus lower limit.

Correct option: B

Final answer

$Fb - Fa$

Worked Solutions **Integration: Definite Integrals and Applications**

Q037 MCQ

When the limits are equal, $\int_a^a f(x)dx$ equals:

Solution

1. There is no interval of accumulation.

Correct option: C

Final answer

0

Q038 MCQ

A reversed definite integral satisfies:

Solution

1. Reversing the limits changes the sign.

Correct option: B

Final answer

negative reversed integral

Worked Solutions **Integration: Definite Integrals and Applications**

Q039 MCQ

To evaluate a definite integral of a product, first:

Solution

1. Then apply the power rule and endpoint substitution.

Correct option: B

Final answer

expand the product

Q040 MCQ

Adjacent integrals with the same integrand can be:

Solution

1. Use interval additivity.

Correct option: B

Final answer

combined over the joined interval

Worked Solutions **Integration: Definite Integrals and Applications**

Q041 MCQ

To use $\int_{\alpha}^{\beta}(x-\alpha)(x-\beta)dx$, the factors must have:

Solution

1. Match each factor to its endpoint.

Correct option: B

Final answer

roots a and b

Q042 MCQ

The identity for the factored integral is:

Solution

1. Use the signed cubic interval length.

Correct option: B

Final answer

$-(b-a)^3/6$

Worked Solutions **Integration: Definite Integrals and Applications**

Q043 MCQ

A minus sign before an integral can be removed by:

Solution

1. $-\int_a^b f = \int_b^a f$.

Correct option: B

Final answer

reversing its limits

Q044 MCQ

In $f(x) = p(x) + \int_a^b f(t)dt$, the definite integral is:

Solution

1. The limits contain no x.

Correct option: B

Final answer

a constant

Worked Solutions **Integration: Definite Integrals and Applications**

Q045 MCQ

The best first step for an integral equation is to let:

Solution

1. Replace the repeated definite integral by one symbol.

Correct option: B

Final answer

$A = integral$

Q046 MCQ

For the warm-up $f(x) = x^2 + \int_0^1 f(t)dt$, the constant integral is:

Solution

1. Solve $A = 1/3 + A$ after substitution.

Correct option: B

Final answer

$-1/3$

Worked Solutions **Integration: Definite Integrals and Applications**

Q047 MCQ

$\frac{d}{dx}\int_a^x f(t)dt$ **equals:**

Solution

1. Apply the variable-upper-limit theorem.

Correct option: B

Final answer

fx

Q048 MCQ

When differentiating an integral, the dummy variable t is replaced by:

Solution

1. The derivative evaluates the integrand at the moving limit.

Correct option: B

Final answer

upper-limit variable

Worked Solutions **Integration: Definite Integrals and Applications**

Q049 MCQ

If $\int_a^a f(t)dt = 0$, this can determine:

Solution

1. Substitute the equal limits into the original equation.

Correct option: B

Final answer

the unknown lower limit a

Q050 MCQ

For $\int_3^x (2t^2 + 3t - 5)dt$, the derivative is:

Solution

1. Use the Fundamental Theorem directly.

Correct option: A

Final answer

$2x^2 + 3x - 5$

Worked Solutions

Integration: Definite Integrals and Applications

Quiz WRITTEN 1 **Concept Quiz · Written**

Find $int(4x^3 - 6x + 2)dx$.

Solution

1. Apply the power rule term by term.

Final answer

$x^4 - 3x^2 + 2x + C$

Quiz WRITTEN 2 **Concept Quiz · Written**

Find $F(x)$ if $F'(x) = 2x - 5$ and $F(2) = 1$.

Solution

1. Integrate and use $F(2) = 1$.

Final answer

$F(x) = x^2 - 5x + 7$

Worked Solutions **Integration: Definite Integrals and Applications**

Quiz WRITTEN 3 Concept Quiz · Written

Evaluate $\int_{-1}^{2}(3x^2 - 4x + 1)dx$.

Solution

1. Use an antiderivative and subtract the endpoint values.

Final answer

6

Quiz WRITTEN 4 Concept Quiz · Written

Find $f(x)$ if $f(x) = x^2 + \int_0^1 f(t)dt$.

Solution

1. Let $A = \int_0{}^1 f(t)dt$; solve $A = 1/3 + A$ after substitution.

Final answer

$f(x) = x^2 - 1/3$

Worked Solutions **Integration: Definite Integrals and Applications**

Quiz MCQ 5 **Concept Quiz · MCQ**

The derivative of $\int_0^x (3t^2+1)dt$ is:

Solution

1. Evaluate the integrand at $t = x$.

Correct option: A

Final answer

$3x^2 + 1$

Quiz MCQ 6 **Concept Quiz · MCQ**

A primitive of $5x^4$ is:

Solution

1. Divide the power coefficient by the new exponent.

Correct option: A

Final answer

$x^5 + C$

Worked Solutions **Integration: Definite Integrals and Applications**

Quiz MCQ 7 **Concept Quiz · MCQ**

For $F'(x) = 4x + 2$ and $F(0) = 3$, $F(x)$ is:

Solution

1. Integrate and apply $F(0) = 3$.

Correct option: A

Final answer

$2x^2 + 2x + 3$

Quiz MCQ 8 **Concept Quiz · MCQ**

$\int_1^3 f(x)dx + \int_3^5 f(x)dx$ equals:

Solution

1. Join adjacent intervals with the same integrand.

Correct option: A

Final answer

integral from 1 to 5

Answer key

Use the question number shown on each page.

Q001 Integrate term by term and add C to each primitive.

Q002 Expand products or powers, collect like powers, then integrate every term.

Q003 Apply linearity and the power rule; retain the constant of integration.

Q004 Expand each product or square, simplify, and integrate term by term.

Q005 $F(x) = x^3 + 4x^2 - x + 3$.

Q006 Integrate the derivative and use the given point to determine C.

Q007 For each item, integrate $F'(x)$, then impose the stated value of F.

Q008 $f(x) = x^3 - x^2 + x - 19$.

Q009 Integrate the given gradient and use the point on the curve to determine the constant.

Q010 Integrate the polynomial gradient and apply the given point condition in each case.

Q011 Find an antiderivative F and calculate $F(b) - F(a)$; equal limits give zero.

Q012 Expand products, integrate, and evaluate the antiderivative at the upper and lower limits.

Q013 Use the Fundamental Theorem; a reversed interval changes the sign and equal limits give 0.

Q014 Use linearity, expand products, then apply the endpoint-substitution rule.

Q015 Scale constants, reverse limits, split intervals, or use the factored-interval identity as appropriate.

Q016 Swap limits when necessary, join adjacent intervals, and normalize a linear factor before using the identity.

Q017 Rewrite each expression until adjacent intervals have matching integrands and compatible limits.

Q018 Scale to $\int_a^b (x-a)(x-b)\,dx = -\frac{(b-a)^3}{6}$.

Q019 Let $A = \int_0^{\,1} f(t)\,dt$; then $A = frac13 + A$, so $A = -frac13$ and $f(x) = x^2 - frac13$.

Q020 Introduce a constant for the definite integral, evaluate it using the proposed $f(t)$, and solve the linear equation.

Q021 Let the definite integral be A, substitute the expression for $f(t)$, solve for A, then recover $f(x)$.

Q022 $2x^2 + 3x - 5$.

Q023 Replace t by x when differentiating, then use equal limits or the given condition to determine a.

Q024 Apply the variable-limit Fundamental Theorem directly to each integrand.

Q025 Differentiate to get $f(x)$, then set the integral to zero at $x = a$ to solve for a.

Q026 B

Q027 B

Q028 B

Q029 A

Q030 B

Q031 C

Q032 A

Q033 B

Q034 B

Q035 B

Q036 B

Q037 C

Q038 B

Q039 B

Q040 B

Q041 B

Q042 B

Q043 B

Q044 B

Q045 B

Q046 B

Q047 B

Q048 B

Q049 B

Q050 A

Quiz WRITTEN 1 $x^4 - 3x^2 + 2x + C$

Quiz WRITTEN 2 $F(x) = x^2 - 5x + 7$

Quiz WRITTEN 3 6

Quiz WRITTEN 4 $f(x) = x^2 - 1/3$

Quiz MCQ 5 A

Quiz MCQ 6 A

Quiz MCQ 7 A

Quiz MCQ 8 A

Concept 5 | Applications of Integration to Geometry

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Formula opener

Area

$$A = \int_a^b (y_{top} - y_{bottom})dx$$

Volume

$$V = \pi \int_a^b y^2 dx$$

Bounds

$$a \leq x \leq b$$

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Worked Solutions **Applications of Integration to Geometry**

Q001 Written

Find the area bounded by $y = 2x^2 + 1$, the x-axis, $x = -2$, and $x = 1$.

Solution

1. The parabola is above the x-axis on the interval.
2. $S = \int_{-2}^{1}(2x^2 + 1)\,dx = 9$.

Final answer

9

Q002 Written

Find the area bounded by $y = x^2 + x - 6$, the x-axis, $x = -2$, and $x = 1$.

Solution

1. The curve is below the x-axis on $[-2, 1]$.
2. Use $S = -\int_{-2}^{1}(x^2 + x - 6)\,dx = \frac{33}{2}$.

Final answer

$$\frac{33}{2}$$

Worked Solutions **Applications of Integration to Geometry**

Q003 **Written**

Find the area bounded by $y = -x^2 + 4$ and the x-axis.

Solution

1. The intercepts are $x = -2, 2$.
2. Integrate the positive curve from -2 to 2.

Final answer

$\frac{32}{3}$

Q004 **Written**

Find the area under $y = -x^2 - 2x + 3$ and above the x-axis.

Solution

1. The roots are -3 and 1.
2. Integrate $-x^2 - 2x + 3$ between the roots.

Final answer

$\frac{32}{3}$

Worked Solutions Applications of Integration to Geometry

Q005 Written

Find the area between $y = x^2 - 2$ and $y = -2x + 1$.

Solution

1. Intersections give $x = -3, 1$; the line is upper.
2. Integrate upper minus lower over $[-3, 1]$.

Final answer
$$\frac{32}{3}$$

Q006 Written

Find the area between $y = x^2 + 2x - 4$ and $y = -x^2 + 2x + 4$.

Solution

1. The intersections are $x = -2, 2$.
2. Use the upper-minus-lower integral $\int_{-2}^{2}(8 - 2x^2)\,dx$.

Final answer
$$\frac{64}{3}$$

Q007 Written

Find the area between $y = -x^2 + 4x$ and $y = x$.

Solution

1. Intersections are $x = 0, 3$.
2. Integrate $-x^2 + 3x$ from 0 to 3.

Final answer

$\frac{9}{2}$

Q008 Written

Find the area between $y = -x^2 + 8$ and $y = 2x$ for $-2 \leq x \leq 3$.

Solution

1. The curves meet at $x = 2$ within the interval.
2. Split the absolute difference at $x = 2$; the two areas are 80/3 and 10/3.

Final answer

30

Worked Solutions **Applications of Integration to Geometry**

Q009 Written

Find the area bounded by $y = x^3 - 5x^2 + 6x$ and the x-axis.

Solution

1. Roots are (0,2,3); the curve changes sign at 2.
2. Add the positive area on $[0, 2]$ to the absolute area on $[2, 3]$.

Final answer

$\frac{37}{12}$

Q010 Written

Find the area bounded by $y = -x^3 + 4x$ and the x-axis.

Solution

1. Roots are (-2,0,2); the two lobes have equal area.
2. Double $\int_0^2 (-x^3 + 4x)\,dx = 4$.

Final answer

8

Worked Solutions **Applications of Integration to Geometry**

Q011 **Written**

Solve the given cubic area exercises by finding every x-intercept and adding the absolute signed pieces.

Solution

1. Factor the cubic first.
2. Mark the sign on each interval before integrating.

Final answer

Use the numbered solution for each given cubic.

Q012 **Written**

Evaluate $\int_{-2}^{1} |x+1|\,dx$.

Solution

1. The sign changes at $x = -1$.
2. Split into two ordinary integrals and add the positive areas.

Final answer

$\frac{5}{2}$

Worked Solutions **Applications of Integration to Geometry**

Q013 **Written**

Evaluate $\int_0^3 |x^2 - 4|\, dx$.

Solution

1. The root in the interval is $x = 2$.
2. Use $-\int_0^{2}(x^2 - 4)\, dx + \int_2^{3}(x^2 - 4)\, dx$.

Final answer

$\frac{23}{3}$

Q014 **Written**

Evaluate $\int_0^2 |x^2 + 3x - 4|\, dx$.

Solution

1. The relevant root is $x = 1$.
2. Reverse the sign on $[0, 1]$, then add the integral on $[1, 2]$.

Final answer

5

Worked Solutions **Applications of Integration to Geometry**

Q015 Written

Find the area bounded by $y = x^2 - x$ and its tangents at $(0, 0)$ and $(2, 2)$.

Solution

1. The tangents are $y = -x$ and $y = 3x - 4$, meeting at $x = 1$.
2. Split the area at $x = 1$ and integrate curve minus tangent.

Final answer

$\frac{3}{2}$

Q016 Written

Find the area bounded by $y = x^3 + x^2 - 3x + 6$ and its tangent at $(1, 5)$.

Solution

1. The tangent is $y = 2x + 3$; the intersections are $x = -3, 1$.
2. Factor the difference as $(x - 1)^2(x + 3)$ and integrate.

Final answer

$\frac{64}{3}$

Worked Solutions **Applications of Integration to Geometry**

Q017 Written

For $y = x^2$, tangents through $(3, 8)$ touch at $x = 2, 4$. Find the enclosed area.

Solution

1. The tangent equations are $y = 4x - 4$ and $y = 8x - 16$.
2. Integrate the two squared gaps on $[2, 3]$ and $[3, 4]$.

Final answer

$$\frac{2}{3}$$

Q018 Written

If $y = kx$ bisects the area under $y = 2x - x^2$ and above the x-axis, find k.

Solution

1. The total area under the parabola is 3.
2. Set the area cut off by $y = kx$ equal to $3/2$, giving $(2 - k)^3 = 3/4$.

Final answer

$$k = 2 - \sqrt[3]{\frac{3}{4}}$$

Worked Solutions **Applications of Integration to Geometry**

Q019 Written

Solve the given bisection exercises for the given parabolas and lines, using the required area ratio.

Solution

1. Find the total parabola area first.
2. Integrate between the line and curve, then impose the stated ratio.

Final answer

Use the numbered solution for each given ratio.

Q020 MCQ

When the graph is below the x-axis, area is:

Solution

1. Reverse the sign.

Correct option: B

Final answer

negative of the integral

Worked Solutions Applications of Integration to Geometry

Q021 MCQ

A bounded area under a curve is found by first checking:

Solution

1. Draw the graph.

Correct option: A

Final answer

the graph and sign

Q022 MCQ

The area under $y = -x^2 + 4$ between its roots is:

Solution

1. Integrate from -2 to 2.

Correct option: C

Final answer

32/3

Worked Solutions
Applications of Integration to Geometry

Q023 MCQ

Between two curves, the integrand is:

Solution

1. Identify the upper graph.

Correct option: B

Final answer

upper minus lower

Q024 MCQ

If curves cross inside the interval, area must be:

Solution

1. The sign of the difference changes.

Correct option: A

Final answer

split at the crossing

Worked Solutions Applications of Integration to Geometry

Q025 MCQ

The area between $y = x^2 - 2$ and $y = -2x + 1$ is:

Solution

1. Use the intersection limits.

Correct option: B

Final answer

32/3

Q026 MCQ

For cubic area, the sign changes at:

Solution

1. Mark all roots.

Correct option: B

Final answer

each x-intercept

Worked Solutions **Applications of Integration to Geometry**

Q027 MCQ

The area for $y = -x^3 + 4x$ is:

Solution

1. The two lobes are equal.

Correct option: C

Final answer

8

Q028 MCQ

A negative lobe contributes:

Solution

1. Area is non-negative.

Correct option: B

Final answer

positive absolute area

Worked Solutions **Applications of Integration to Geometry**

Q029 MCQ

For an absolute-value integral, first locate:

Solution

1. Roots set the sign intervals.

Correct option: A

Final answer

the roots

Q030 MCQ

The value of $\int_0^2 |x^2+3x-4| dx$ is:

Solution

1. Split at $x = 1$.

Correct option: C

Final answer

5

Worked Solutions

Applications of Integration to Geometry

Q031 MCQ

If f changes sign at c, split the integral:

Solution

1. The absolute value changes form.

Correct option: A

Final answer

atc

Q032 MCQ

A tangent line at x=a has slope:

Solution

1. Differentiate the curve.

Correct option: B

Final answer

$f'a$

Worked Solutions **Applications of Integration to Geometry**

Q033 MCQ

For two tangent lines, split at their:

Solution

1. The upper tangent changes there.

Correct option: A

Final answer

intersection x-coordinate

Q034 MCQ

The area in the given parabola-tangent Warm Up is:

Solution

1. Integrate the two gaps.

Correct option: C

Final answer

3/2

Worked Solutions **Applications of Integration to Geometry**

Q035 MCQ

Bisection means each part has:

Solution

1. Set each part to half the total.

Correct option: A

Final answer

the same area

Q036 MCQ

For $y=2x-x^2$, the total area above the x-axis is:

Solution

1. Integrate from 0 to 2.

Correct option: C

Final answer

3

Worked Solutions Applications of Integration to Geometry

Q037 MCQ

The line y=kx must satisfy the stated:

Solution

1. Use the given bisection condition.

Correct option: A

Final answer

area ratio

Q038 MCQ

In a bisection problem, the cut-off area is:

Solution

1. Set the two parts equal.

Correct option: B

Final answer

half the total area

Worked Solutions **Applications of Integration to Geometry**

Quiz MCQ 1 Concept Quiz · MCQ

The area under $y = -x^2 + 4$ and above the x-axis is:

Solution

1. Integrate from -2 to 2.

Correct option: C

Final answer

$\frac{32}{3}$

Quiz MCQ 2 Concept Quiz · MCQ

The area between $y = x^2 - 2$ and $y = -2x + 1$ is:

Solution

1. Use upper minus lower from -3 to 1.

Correct option: B

Final answer

$\frac{32}{3}$

Worked Solutions **Applications of Integration to Geometry**

Quiz MCQ 3 **Concept Quiz · MCQ**

Evaluate $\int_{-2}^{1} |x+1|\, dx$:

Solution

1. Split at $x = -1$.

Correct option: B

Final answer

$\frac{5}{2}$

Quiz MCQ 4 **Concept Quiz · MCQ**

The area enclosed by $y = x^2$ and tangents through $(3, 8)$ is:

Solution

1. Integrate the two squared gaps.

Correct option: C

Final answer

$\frac{2}{3}$

Worked Solutions **Applications of Integration to Geometry**

Quiz WRITTEN 5 **Concept Quiz · Written**

Find the area bounded by $y = x^3 - 5x^2 + 6x$ and the x-axis.

Solution

1. Use roots (0,2,3) and split at 2.

Final answer

$\frac{37}{12}$

Quiz WRITTEN 6 **Concept Quiz · Written**

Evaluate $\int_0^2 |x^2 + 3x - 4|\, dx$.

Solution

1. Split at the root $x = 1$.

Final answer

5

Worked Solutions **Applications of Integration to Geometry**

Quiz WRITTEN 7 Concept Quiz · Written

Find the area bounded by $y = x^2 - x$ and its tangents at $(0, 0)$ and $(2, 2)$.

Solution

1. Find both tangents and split at their intersection.

Final answer

$\frac{3}{2}$

Quiz WRITTEN 8 Concept Quiz · Written

If $y = kx$ bisects the area under $y = 2x - x^2$, find k.

Solution

1. Set the cut-off area equal to half of 3.

Final answer

$2 - \sqrt[3]{\frac{3}{4}}$

Answer key

Use the question number shown on each page.

Q001 9

Q002 $\frac{33}{2}$

Q003 $\frac{32}{3}$

Q004 $\frac{32}{3}$

Q005 $\frac{32}{3}$

Q006 $\frac{64}{3}$

Q007 $\frac{9}{2}$

Q008 30

Q009 $\frac{37}{12}$

Q010 8

Q011 Use the numbered solution for each given cubic.

Q012 $\frac{5}{2}$

Q013 $\frac{23}{3}$

Q014 5

Q015 $\frac{3}{2}$

Q016 $\frac{64}{3}$

Q017 $\frac{2}{3}$

Q018 $k = 2 - \sqrt[3]{\frac{3}{4}}$

Q019 Use the numbered solution for each given ratio.

Q020 B

Q021 A

Q022 C

Q023 B

Q024 A

Q025 B

Q026 B

Q027 C

Q028 B

Q029 A

Q030 C

Q031 A

Q032 B

Q033 A

Q034 C

Q035 A

Q036 C

Q037 A

Q038 B

Quiz MCQ 1 C

Quiz MCQ 2 B

Quiz MCQ 3 B

Quiz MCQ 4 C

Quiz WRITTEN 5 $\frac{37}{12}$

Quiz WRITTEN 6 5

Quiz WRITTEN 7 $\frac{3}{2}$

Quiz WRITTEN 8 $2 - \sqrt[3]{\frac{3}{4}}$